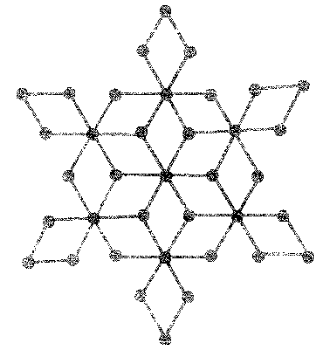
GEOMETRY EXERCISE

SECTION A

1. Inthe adjoiningrangoli design each ofthe four sided figures isarhombus and thedistancebetweenany two dots is 1unit.The totalareaofthe designis



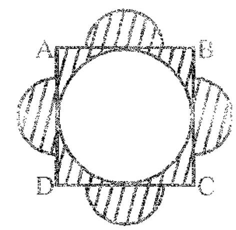
(1) 363sq.units

(2) 93sq.units

(3) 243sq.units

(4) 183sq.units

2. Intheadjoiningfigure,ABCDisasquareofside4units.Semicirclesaredrawnoutsidethesquareswithdiameter 2units asshown.Theareaoftheshadedportioninsquareunitsis



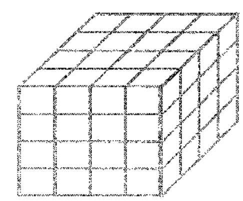
(1)8

(2)16

(3) 16-21t

(4) 8- *n*

3. Acube ofedge 4cm is paintedexternally withred colar. If isthen cutintoonecm cubes.How many ofthese donothaveredpaintonanyface?



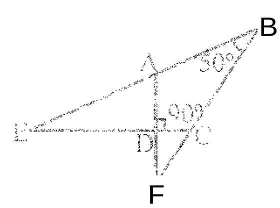
(1)4

(2)8

(3) 56

(4)16

4. IntheadjacentfigureBAandBCareproducedtomeetCDandADproducedisEandF.Then *LAED+LCFD*is



(1) 80°

(2) 50°

(3) 40°

(4) 160°

5. InanisoscelestriangleABC,AC=BC,*LBAC*isbisectedbyADwhereDliesonBC.Itisfoundthat AD=AB. Then*LACB* equals

(1)72°

(2)54°

(3)36°

(4) noneofthese

6. ABCisarightangledtrianglewithL'.'.BAC=9OQ.AHisdrawnperpendiculartoBCwhereHliesonBC. IfAB=60andAC=80,thenHH=

(1)36

(2)32

(3)24

(4)30

7. Threeidenticalrectanglesareoverlappingasinthediagram.Thelengthandbreadthofeachrectangleare respectively2007cmand10cm.Theareaofeachoftheshadedsquareportionsis16cm2.

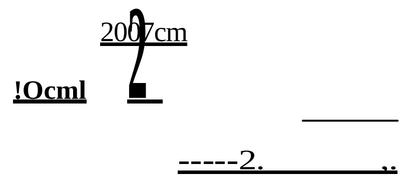
Theperimeteroftheouterboundaryofthefigureincmis

(1)10070

(2)12070

(3)14070

(4)11070



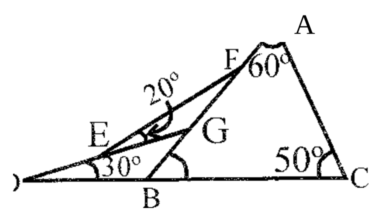
8. IntheadjoiningfigureA=60°,C=50°.L'.'.BDG=30°,L'.'.GEF=20°,Then

(1) EG=2FG

(2) EG>FG

(3) EG=FG

(4) EG<FG



9. Thelengthsofthealtitudesofatriangleareintheratio1:2 :3.Then

(1) oneangleofthetrianglemustbe60°

(2) thetriangleisarightangledtriangle

(3) thetriangleisanobtuseangledtriangle

(4) suchatriangledoesnotexist

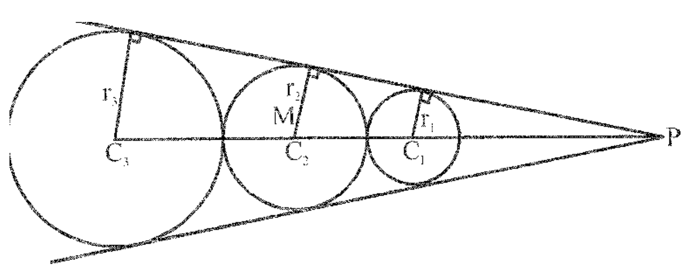
10. ThreecirclesC1,C2,C3withradiir1,r2,r3wherer1<r2<r3areplacedasinthefigure.Thenr2

(1) .Jr3-f1

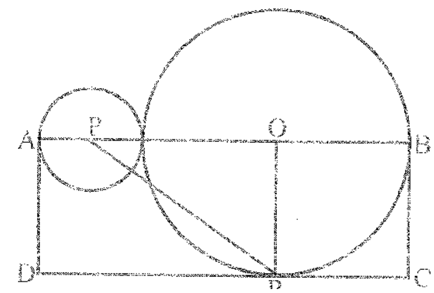
(2) .Jr3+r1

(3) $\sqrt{r_3^2 - r_1^2}$

(4) $\sqrt{r_3 r_1}$



Twocirclesofdifferentradiitouch externally.APQBisthelineofdiarneter as shownin the diagram. AD, BC and DRC are tangents so that the areaofthe rectangieABCDisJ6cm^2.The areaofthe trianglePQR iscm^2.



(1) 4cm^2

(2) 6crn^2

(3) 9cm^2

(4) 14cm^2

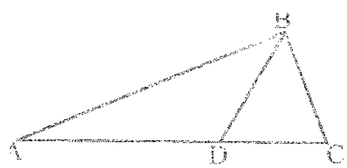
12. Twoofthealtitudesofthe scalenetriangleABChavelengths 4and12.ifthelengthofthethirdaltitudeis also aninteger,thenits greatestvaluecanbe______________________

(1)8 (2) 5 (3) 7 (4)6

13. Whichoneofthefollowinganglescannotbeconstructedusinganunmarkedrulerandcompassonly?

(1)75° (2)90° (3}50° (4) $22\frac{1}{2}°$

14. IntriangleABC,BDbisectangieB.lfm,::'.'.A= $\frac{2}{3}$ rnBandm/B ° 3mL'.Cthemm:'.'.'.BDCis:-



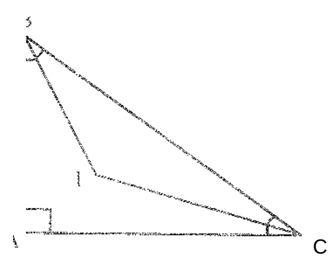
(1) 7S°

(2) 105°

(3) 90°

(4) 120°

15. Intheadjoiningfigure,BACisrightanfdedatA;mL'.B=*mL'.C.*ThebisectorsofanglesBandCmeetat I.ThenmL'.BICis:-



(1)135°

(2)115°

(3i100°

(4)90°

16. ThenumberofisoscelestrianglesinwhichoneE,ngleis4i.irnesanotherangleIs:•

(1)2 (2)l (3)infinitelymany (4)4

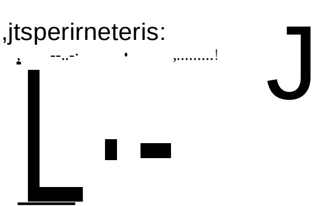
17. ABCDisasquare.Pisapointinitsinieiiorsuchthat,'1PABisequilateral.Thenmeasureof,ePCBis:-

1) 45° (2)30° (3)60° 4) 75°

18. Arhombushasonediagonaldoubletheother.Ifthereaofrh.::irnbusiskcm^{2}ihentheleqnth itssideis

{1\--·cm 4 (2) ------cm 4 (3) $\sqrt{\frac{5k}{4}}$cm (4)------- cm 4

19. ThEadjoiningfigureisformedbyt,vosquares.Thesideofeachsquareisawholenumber. Htheareaofthefigureis58squarecenti-rneters,jtsperirneteris:



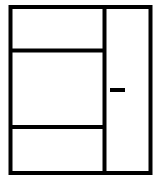
(1) 58cn1

(2) 34crn

(3) 29crn

(4) 1?0 cm

io. Arectanglewithperimeter44ispartitionedinto5congruentrectangles,asindicatedinthediagram.The perimeterofeachofthecongruentrectangleis :-



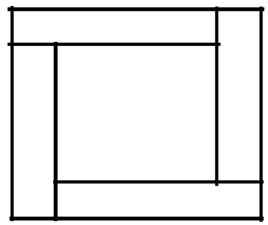
(1)10

(2)20

(3)24

(4)48

H. Thesmallsquare issurroundedby fourcongruentrectangles asshown,to forma largesquare. If theperimeter ofeachrectangleis 16,theareaofthelargesquareis:-



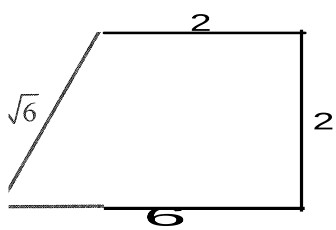
(1) 64

(2)256

(3)32

(4)224

22. Theareaofthequadrilateralshowninthefigureis:-



(1)6

(2)2+6

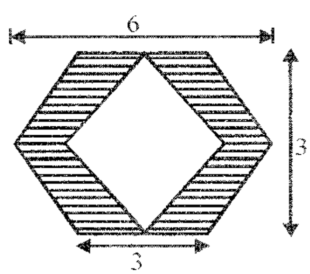
(3)2+8

(4)12

23. If all thediagonalsof aregular hexagon aredrawn, then number ofpointsof intersection,notcounting the corners of the hexagon is :-

(1)6 (2)13 (3) 7 (4)12

24. Theareaoftheshadedregioninthediagramis:-



(1) 9

(2)32

(3)18

(4)63-32

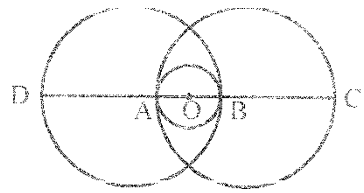
25. The altitude drawnto the base ofanisosceles triangleisoflength8 andtheperimeteris 32.The area ofthe triangle is :-

(1)32 (2)40 (3)48 (4)56

26. Acircleisadded totheequallyspaced gridalongside.Thelargest numberof dotsthatthecirclecan passthrough is :-

(1)4 (2)6 (3)8 (4)10

27. D,A,O,BandCarepoints onaline.Circlesare drawnwith0,A andBascentresas shown. Iftheradiusof the smallestcircleis 1cm,thenthe lengthCD is :-



(1) 3cm

(2) 4cm

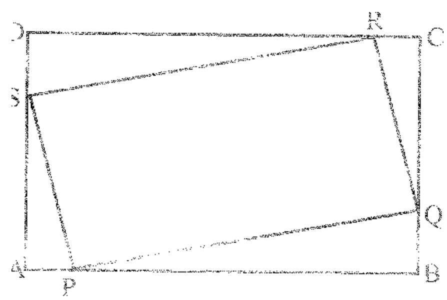
(3) 5cm

(4) 6cm

28. In thetriangle ABC,Disapoint on the line segmentBCsuch that AD=BD =CD.The measure ofangle BAC is:-

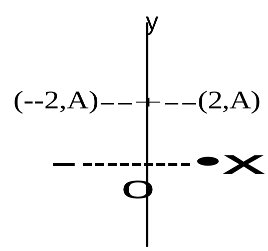
(1)60°　(2)75°　(3)90°　(4)120°

29. Point P,Q,RandSdividethesidesofa rectangle ABCD in the ratio1: 2,asshowninthe figure. What fraction ofareaoftherectangleis theareaoftheparallelogramPQRS?



(1) $\frac{2}{5}$　(2) $\frac{3}{5}$　(3) $\frac{4}{9}$　(4) $\frac{5}{9}$

30. SupposeamirrorstandsontheY-axisdrawnonagraphsheet.Thenthe reflectionofthepoint(-2,4)themirror willbe



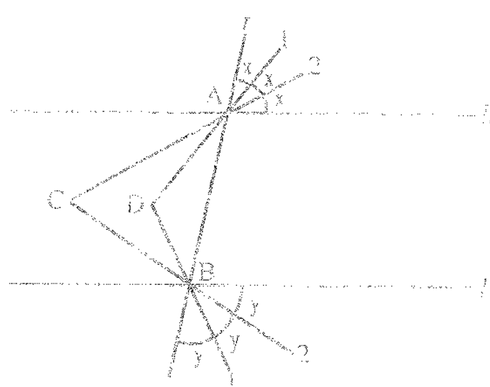
(1)(2,4)

(2)(2,-4)

(3)(4,2)

(4)(4,-2)

31. Intheadjoiningfigure l_1 and l_2 areparallellines. t isatransversalwhichcuts l_1, l_2 atA,Brespectively. Theangles atA,Baretrisected.ThemeasuresoftheanglesACBandADBarerespectively

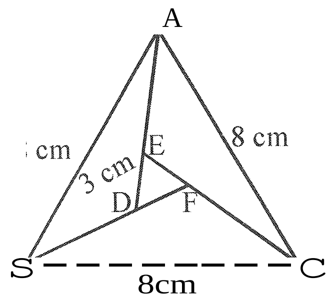


(1)60°,120°　(2)120°,60°　(3)x-Y,x+Y　(4)2(x-y),2(x+y)

2. ABCisan isosceles triangle with AB=AC=2008cm. ADC isdrawn asan equilateraltriangle on ACoutside ABC.ADisparalleltoBC.ThebisectorofDmeetsABinE,say.ThenBEisequalto

(1) 1004cm (2) 2008cm (3) 0 (4) 502cm

3. In theadjoining figure ABC, DEFareequilateraltriangles.AB=8cmand DE= 3cm.Then the possible value ofAE +BD +CF is

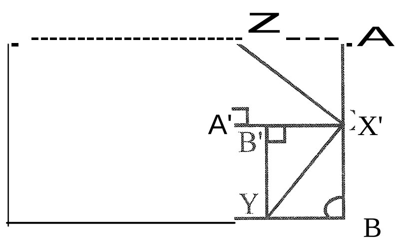


(1)6.9cm (2)7.1cm (3) 5.2cm (4)8.3cm

14. Themeasureofone oftheanglesofarighttriangleis five times thatofasecondangle.Thenthepossibility ofthe second largest angle is

(1)72° (2)75° (3) 72°or75° (4) Noneofthese

15. Arectangularsheetofpaperisfoldedsothatthecorners A,BgotoA',B'asinthefigure.Then*LZXY*is



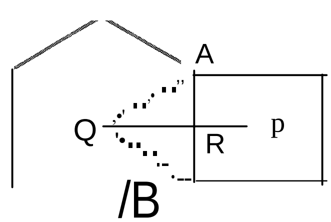
(1) anacuteangle

(2) anobtuseangle

(3) arightangle

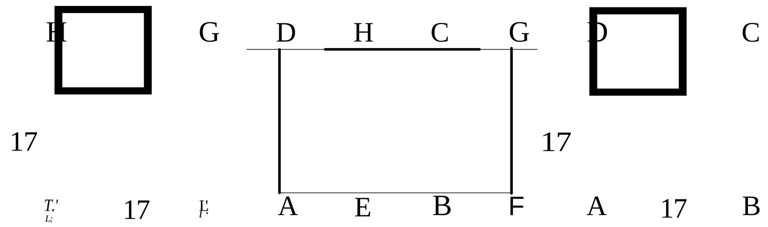
(4) avariableangle dependingonthepoint X

J6. OnalinesegmentAB=2008cmasquareandaregularhexagonaredrawnas showninthediagram.The distancebetweentheir centres P,Q,in cm is



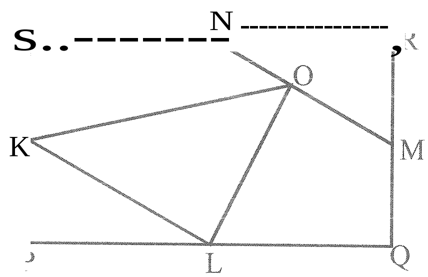
(1)10043 (2)1004(13t1) (3)20083 (4)1004h/3-l)

37. Two sqaures17cmx17cm overlapto formarectangle17cmx30cm. Theareaofthe overlappingregion is



(1)289 (2)68 (3)510 (4)85

38. Inthe figure,PQRSisarectangleofarea 2011squareunits.K, L,M,Nare themidpoints oftherespective sides.0is themidpointofMN.Thearea ofthetriangleOKL is equalto(insquareunits)



(1) $\frac{2011}{5}$ (2) $\frac{2(2011)}{5}$ (3) $\frac{2011}{4}$ (4) $\frac{3(2011)}{8}$

39. PQRS isaparallelogram. MPandNPdivideLSPQ into three equalparts (LMPQ>LNPQ) andMQ andNQ divideLRQPinto3equalparts(LMQP>LNQP).Ifk(LPNQ)=(LPMQ)thenk=

(1) $\frac{1}{2}$ (2)1 (3) $\frac{3}{2}$ (4) $\frac{1}{3}$

40. ApointPinsidearectangleABCDisjoinedtotheangularpoints.Then

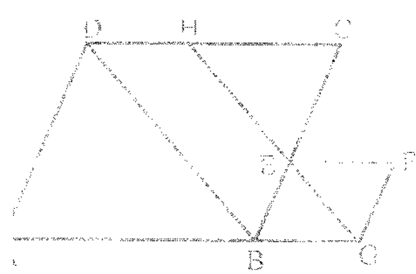
(1) Sumoftheareasoftwoofthetrianglessoformedisequaltothesumoftheothertwo

(2) Thesumoftheareasofthetrianglessoformedisawholenumberwhatevermaybethe dimensions of the rectangle.

(3) Thesumoftheareasofapairofoppositetrianglesisgreaterthanhalftheareaoftherectangle.

(4) Noneofthese

41. IntheadjoiningfigureABCDandBGFEarerhombus.AB=10cm,GF=3cm.GEmeetsDCatH. *LA*=60°.TheperimeterofABEHDis(incm)



(1)47 (2)40 (3) 39 (4) 33

12. Theperimeterofanisoscelesrightangledtriangleis2012.Itsareais

(1)2012(3-2) (2)$(1006)^2$(3-22)

(3)$(2012)^2$ (4)$(1006)^2$

13. Tworegularpolygons ofsamenumberofsideshavesides40cm and 9cmin legnth.Thelengths ofthe side ofanotherregularpolygonofthesamenumberofsides andwhoseareais equaltothesum oftheareas of thegiven polygons is (in cm)

(1)49 (2)31 (3)41 (4)360

i-4. AXand**BX**aretwoadjacentsidesofaregularpolygon.If LABX=$-\frac{1}{3}$*LAXB*,thenthenumberofsidesof thepolygonis

(1)6 (2)7 (3)9 (4) 5

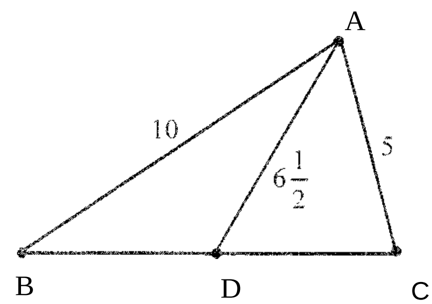
i-5. Arectangularblockofdimensions16x10x8unitsispainted.Itiscutintocubesofdimensions 1x1x1.The number ofcubeswhich are not painted at all is

(1)945 (2)672 (3)812 (4) 796

4-6. Theradiusofacircle is increasedby 4unitsandtheratioofthe areasoftheoriginalandtheincreasedcircle is 4: 9.Theradius ofthe originalcircleis

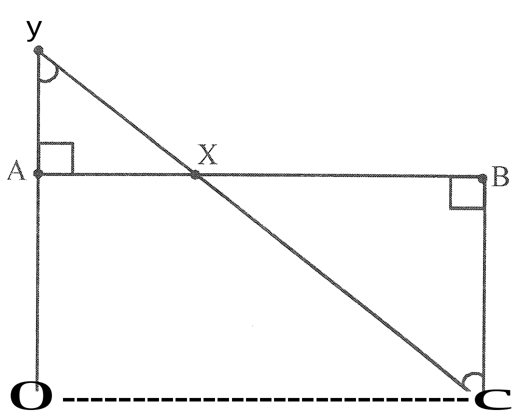
(1)6 (2)4 (3)12 (4) 8

4-7. Twosidesofatriangleare10cmand5cminlengthandthelengthofthemediantothethirdsideis $6\frac{!}{2}$cm.Theareaofthetriangleis6,cm^2.Thevalueofxis



(1)12 (2)13 (3)14 (4)15

48. ABCDisarectangle. Through Cavariablelineisdrawn soastocut ABat Xand DAproduced atY.Then BX xDYis



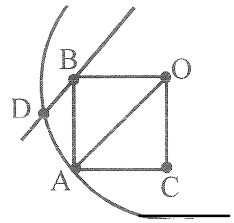
(1) TwicetheareaoftherectangleABCD

(2) EqualtotheareaoftherectangleABCD

(3) AvariablequantitywhichliesbetweentheareaofrectangleABCDandtwicetheareaoftherectangle ABCD

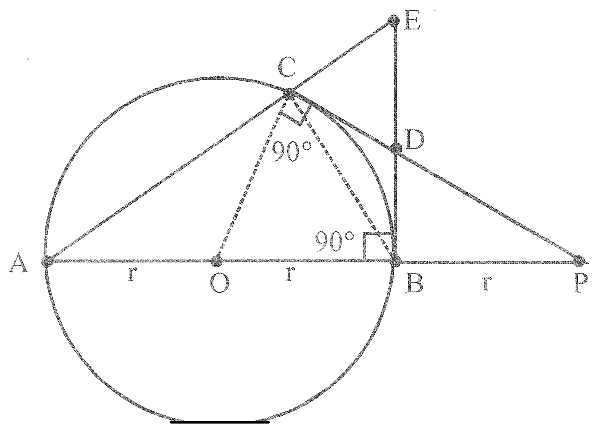
(4) AlwaysaconstantlessthantheareaofrectangleABCD

49. In the adjoining figure, 0 is the centre of the circle. ACOB is a square with A on the circle. Through B a line parallel to OA is drawn to cut the circle at D nearer to A. Then $\angle BOD$=



(1) 20° (2) 18° (3) 15° (4) $22\frac{1}{2}°$

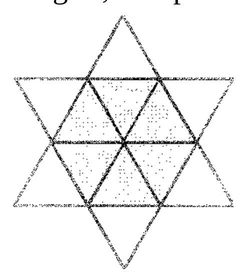
50. In the figure below, AB is a diameter of a circle. AB is produced to P such that BP = radius of the circle. PC is a tangent to the circle. The tangent at B and AC produced cut at E. Then $\triangle$CDE is



(1) isosceles with EC=ED (2) isosceles with EC=CD

(3) equilateral (4) a scalene triangle

51. The star shown in the picture is made by fitting together 12 congruent equilateral triangles,. The perimeter of the star is 36cm. What is the perimeter of the grey hexagon?



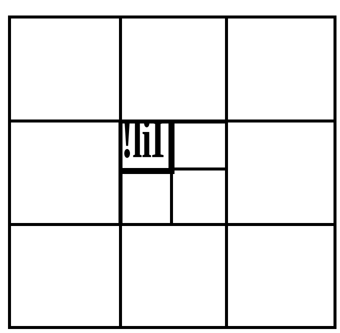
(1) 6cm

(2) 12 cm

(3) 18 cm

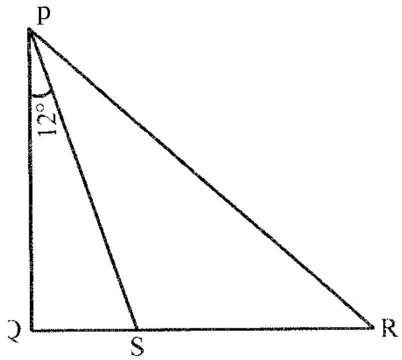
(4) 24cm

52. In the picture the large square has an area of 1. What is the area of the small black square?



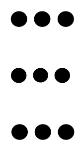
(1) $\frac{1}{100}$ (2) $\frac{1}{300}$ (3) $\frac{1}{600}$ (4) $\frac{1}{900}$

.3. InthediagramQSRisa straightline.LQPS=12°andPQ=PS=RS.HowbigisLQPR?



(1)36° (2)42° (3)54° (4)60°

i4. What isthe minimum number of dotsthat must betakenaway from thepicture sothat nothreeof the remaining dots lie ona straightline?

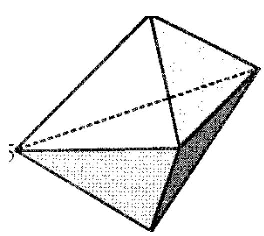


(1) 1 (2)2 (3) 3 (4)4

55. Nickmeasured all 6angles intwotriangles. Oneof thetriangleswasacuteangled and the otherobtuseangled. He noted four oftheangles to be: 120°, 80°, 55° und 10°. What is the size ofthe smallest angle in the acute angled triangle?

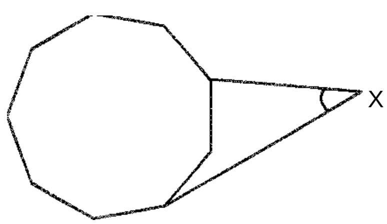
(1)45° (2)10° (3)5° (4)55°

56. In thediagram thereis anobjectwith6triangularfaces.Oneach comer thereis anumber(two are shown). Thesumofthenumbersonthecornersofeachfaceis thesame.Whatisthesumofall5numbers?



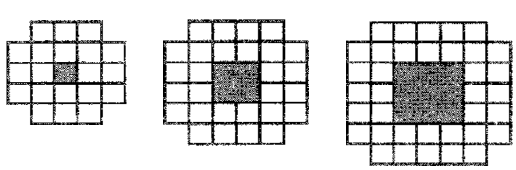
(1)9 (2)12 (3)17 (4)18

57. Thediagramshowsaregularnonagon.WhatisthesizeoftheanglemarkedX?



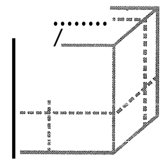
(1)45° (2)50° (3)55° (4)60°

58. A pattern is made out ofwhite, square tiles. Thefirst three patterns are shown. How many tileswill be needed for the tenthpattern?



(1)76 (2)80 (3)84 (4) 92

59. Acubeis cutinthreedirectionsas shown,to produceeightcuboids(eachcutisparalleltooneofthefaces of thecube). What isthe ratio of the total surface area oftheeight cuboids to the surface area ofthe original cube?



(1)1:1 (2)4:3 (3)3:2 (4)2:1

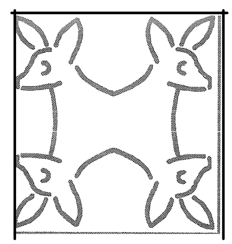
60. In thequadrilateral PQRSPQ = 2006. QR=2008. RS=2007 und SP= 2009. At whichcorners mustthe interior angle definitely be smaller than 180"?

(1)P.Q.R (2)Q.R.S (3)P.Q.S (4)P.R.S

61. IntriangleABC theinteriorangleB equals20°andC40°.ThelengthoftheanglebisectorthroughA is 2. Whatis the differenceofthe side lengths ofBC and AB?

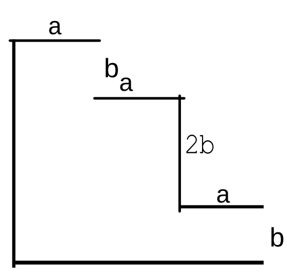
(1) 1 (2)1.5 (3)2 (4)4

62. Howmanylinesofsymmetrydoesthisfigurehave?



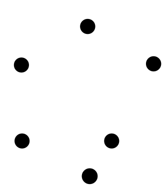
(1)0 (2)1 (3)2 (4)4

63. Theperimeterofthefigurepicturedontherightis..............



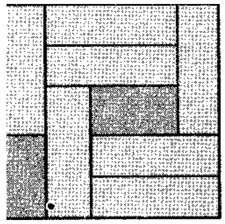
(1)3a+86 (2)6a+46 (3)•6a+66 (4)6a+86

64. Pratapdrawsthesixcomerpointsofaregularhexagonandthenconnectssomeofthemtoobtainageometric figure. Which of thefollowing figurescannot begenerated?



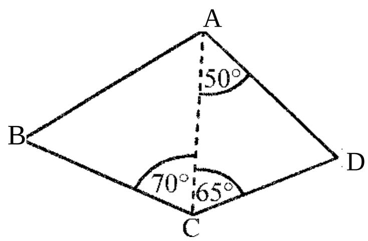
(1)trapezium {2)rightangledtriangle (3)square (4)kite

65. In the box are sevenblocks.It is possibletoslide theblocks aroundso thatanotherblock canbe addedto the box. Whatis theminimumnumberotblocks thatmustbemoved?



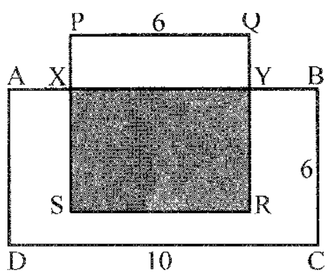
(1)2 (2) 3 (3)4 (4) 5

66. InthequadrilateralABCDAD=BC,*LDAC*=50°,*LDCA*=65°and*LACB*=70°.HowbigisLABC?



(1)50° (2)55° (3)60° (4)65°

67. InthefigureABCDisarectangleandPQRSasquare.Theareaofthegreypartishalfasbigasthearea ofABCD. How long is the side PX?

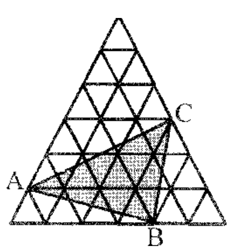


(1) 1 (2)1.5 (3)2 (4)2.5

68. ACamel who isinterestedin geometryhasa collectionof1x1x1 dice.Each diehasacertain colour. It wants to makea3x3x3cubeoutofthe dice so thatsmall dicethatmeetatthevery leaston one comerarealways ofadifferentcolour.Whatis the smallestamountofcoloursitneedstouse?

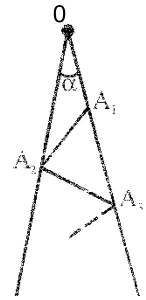
(1)6 (2) 8 (3)9 (4)12

69. Thebigequilateraltriangleconsistsof 36smallequilateraltriangleswhicheachhaveanareaof 1cm^2. Determine the area ofABC.



(1)11crn^2 (2)12cm^2 (3)15cm^2 (4)9cm^2

70. Inthefigure$a$=7°.AlllinesOAAAAA areequallylong.Whatisthemaximumnumberoflinesthat canbedrawninthiswayifnotwolinesareallowedtointersecteachother?

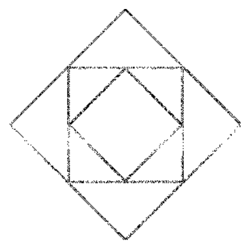


(1)10 (2)11 (3)12 (4)13

71. Eshahas3tetrahedraand5dice.Howmanyfacesdotheseeightobjectshavealtogether?

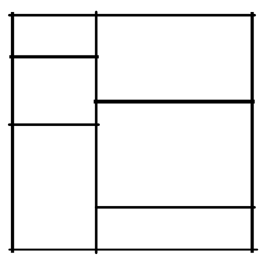
(1)42 (2)48 (3}50 (4)52

72. Inthepicturewecanseethreesquares.Thecornersofthemiddlesquareare onthemidpointsofthesides of thelargersquare, andthecornersof thesmallersquareareonthe midpoints of thesidesof the middlesquare. Theareaofthesmallsquareis6cm^2.Whatistheareaofthebigsquare?



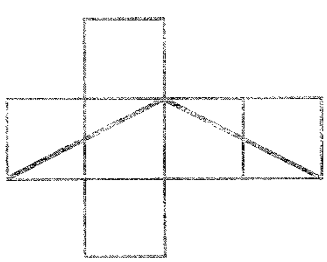
(1)24cm^2 (2)18cm^2 (3)15cm^2 (4)12cm^2

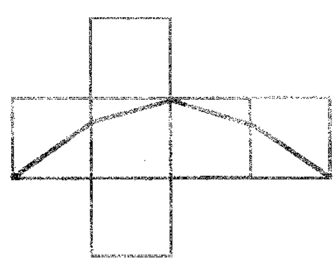
73. Asquarepieceofpaperis cutintosixrectangularpiecesas shownontheright.Thesumoftheperimeters ofthesixpiecesis 120 cm.Howbigistheareaofthesquare?

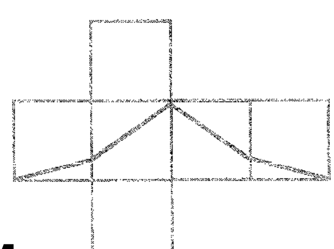


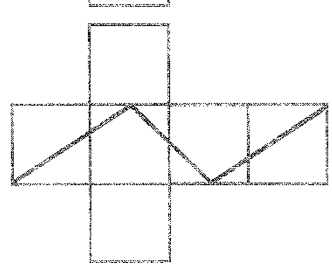
(1)48cm^2 (2)64 cm^2 (3)110.25cm^2 (4)144cm^2

74. Thedarkline halvesthesurfaceareaof thediceshownon theright. Whichdrawingcould represent thenet ofthe die?

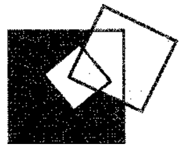
(1) 

(2) 

(3) 

(4) 

75. Ontheinsideofasquarewithsidelength7cmanothersquareisdrawnwithsidelength3cm.Thenathird square with side length 5cmisdrawn so that itcutsthe first twoas shown in the picture onthe right. How big is the differencebetween the black area and thegrey area?



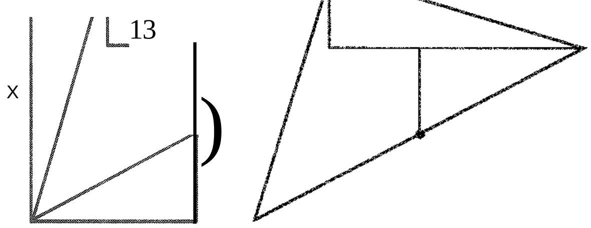
(1)0cm^2 (2)10cm^2 (3)11cm^2 (4)15cm^2

76. InaconvexquadrilateralABCDwithAB=AC,thefollowingholdstrue:LBAD=80°,LABC=75°, *LADC*=65°.Howbigis*LBDC?*(Note:Inaconvexquadrilateralallinternalanglesarelessthan180°.)

(1)10° (2)15° (3)20° (4)30°

77. Thefigureconsistsoftworectangles.Two side lengths aremarked:11and13.The figureis cutinto three parts alongthetwo lines drawninside.These canbeputtogethertomakethetriangleshownontheright. How long is theside marked x?

lli)



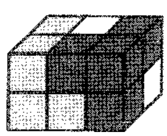
(1)36 (2)37 (3)38 (4)39

78. Each ofthenine pathsina parkare100mlong. Annawants towalkfromAto Bwithout using thesame path twice. How long the longest path she can choose?

B
6
f\/\A

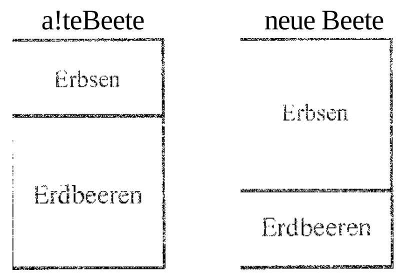
(1)900m (2)800m (3)700m (4)500m

79. Acuboidconsistsof threebuildingblocks.Eachbuilding blockhasadifferentcolourand ismade up of4cubes. What does the white building block look like?



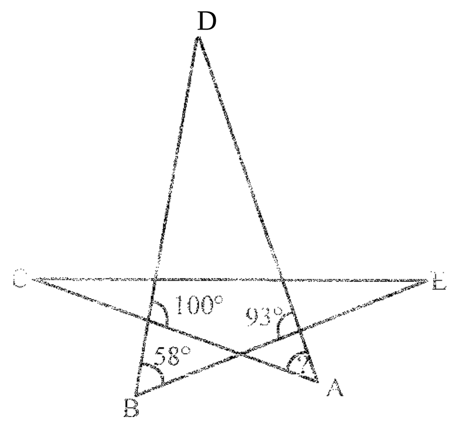
(l)rrf:E (2)@ 13) (4)cill

80. Ms. Green plants peas("Erbsen") andstrawberries ("Erdbeeren") only in her garden. Thisyearshe haschanged her pea·-bed intoasquare-shaped bed byincreasing oneside by 3m,Bydoing thisherstrawberry-bed became 15m²smaller.Whatareadidthepea-bedhavebefore?



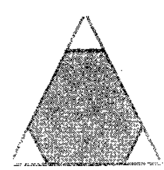
{1)5m² (2)9m² (3)10m² (4) 15m²

SL Thediagramshowsafive-pointedstar.HowbigistheangleA?



(1)35° (2)42° (3)51° (4)65°

82. Three equally sized equilateral triangles are cut from the vertices of a large equi!aterai triangle of side length 6cm. The three little triangles togetherhave the sameperimeter as the remaining grey hexagon. Whatis the side-lengthof oneside ofone small triangle?

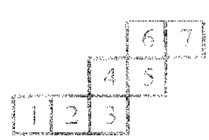


(1)1cm (2)1.2cm (3)1.25cm (4)1.5cm

83. Initially thesidelength ofatalking magicsquare is8cm.Every time it speaksthetrulh itssidesee1ch decrease by 2cm.If iliesits perimeter doubles.It saysfoursentences. twoofwhich aretrueand twoaxefalse, in which order is unknown.Whatis thebiggestpossibleperimeterit canhave after those foure;entences·>

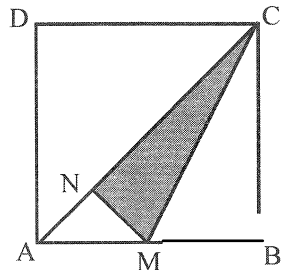
CZ)80 (3)88 (4-)112

84. 'Thediagrarr1showsthe7positionsl,2,3,4,5,6,7ofthebott.ornsideofadievv'hichisrolledaroundits edgeinthisorder.Whichtwocftheseposilionsweretakenupbythesamefaceofthedie?



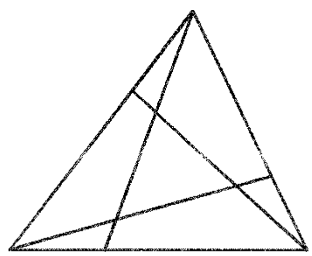
(1)1and7 (2)1and6 (3)1cin<l5 /4)2and7

85. InasquareABCDMisthemidpointofAB.MNisperpenticulartoAC.Determinetheratiooftheareaof thegreytriangletotheareaofthesquareABCD.



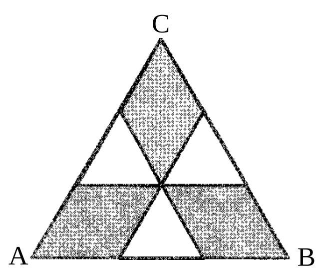
(1)1:6 (2)1:5 (3)7:36 (4)3:16

86. Three lines disect a big triangle into four triangles and three quadrilaterals. The sum ofthe perimeters ofthe threequadrialterals is 25 cm. Thesumoftheperimetersofthefour trianglesis20 cm. Theperimeterofthe bigtriangleis 19cm.Howbigis the sum ofthelengthsofthethreedissectinglines?



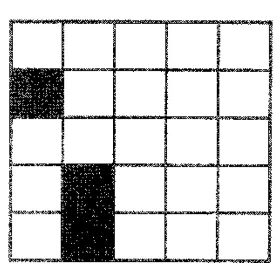
(1)11cm (2)12cm (3)13cm (4)15cm

87. Triangle ABCisequilateral and hasarea 9. Thedividing lines are parallel to thesides, and divide thesides into threeequallengths.Whatistheareaofthegreyshadedpartofthetriangle?



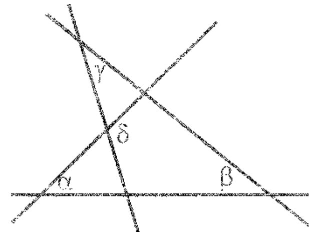
(1) 1 (2)4 (3) 5 (4)6

88. Suhaniplays'sinktheshiplwithafriend,ona5x5grid.Shehasalreadydrawnina1xIshipanda 2 x2 ship(asshown in the picture). Shemust alsodrawa (rectangular) 3x1 ship.Ships may beneither directly nordiagonallyadjacenttoeachother.Howmanypossiblepositionsarethereforthe3x1ship?



(1)5 (2) 6 (3}'/ (4)8

89. Inthediagrampictured,a=55°,f3=40°andy=35°.Howbigis8?



(1)105° (2)120° (3)125° (4)130°

90. The perimeterofatrapezium is5,and the sidelengths arewhole numbers. How many degrees dothesmallest angles measure?

(1)30°and30° (2)60°and60°

(3)45°and45° (4)30°and60°

91. The five shapespicturedwere cutoutofpaper.Fourofthemcanbe foldedtoform acube.Forwhichshape is this not possible.

(l)c=M=i (2) (3) (4)

Shapel Shape2 Shape3 Shape4

92. Vikram stacked !xi cubes on the squares of a 4 x4 grid. The diagram shows the number of cubes that were stackedontop ofeachotheraboveeachsquare.WhatwillVikramseeifhe looks fromtheback(hinten)at the tower?

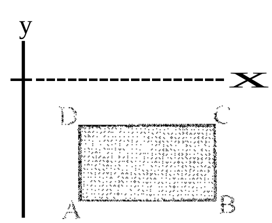
HINTEN

4	2	3	2
3	[illegible]	l	2
2	1	3	1
1	2	1	2

VORNE

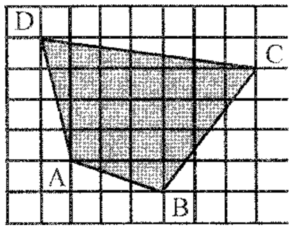
(1)8:HJ (2) (3)Ed (4)h

93. ThesidesoftherectangleABCDareparallelto theco-ordinate axes. The rectangle ispositionedbelowthe x-axisandtotherightofthey-axis,asshowninthe'picture.Theco-ordinatesofthepoints A,B,C,Darewholenumbers.Foreachofthepointswecalculatethevalueof(yco-ordinate) -r(xco-ordinate).Whichofthepointswillgivethesmallestvalue?



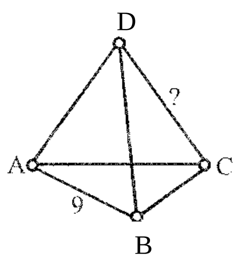
(1)A (2)B (3)C (4)D

)4. Onu sheetofpaperagridisdrawnsuchthateachofthesquareshassides2cmlong.Howbig isthe area ofthegreyshadedquadrilateralABCD?



(1)96cm^2 (2)84cm^2 (3)76cm^2 (4)88cm^2

:)5.Each of the4 verticesand 6 edgesof a tetrahedron islabelled with oneof the numbers1,2, 3,4,5, 6,7,8,9 and 11.(The number10isleft out). Each number isonly used once. The numberon each edge isthesum ofthenumbers onthe two vertices which are connectedby thatedge. The edge AB has the number9. With which numberis the edge CD labelled?



(1)4 (2) 5 (3)6 (4)8

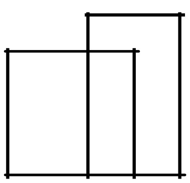
96. Howmanyquadrilateralsofanysizearetobefoundinthediagrampictured.

(1) 0

(2}1

(3) 2

(4)4



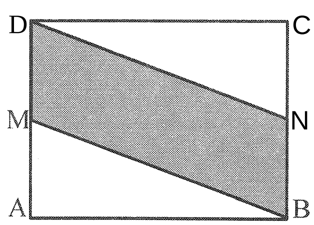
97. TheareaofrectangleABCDinthediagramis10.MandNarethemidpointsofthesidesADandBC. respectively.How big is the area ofthequadrilateralMBND?

(1)0-5

(2)5

(3)2-5

(4)7.5



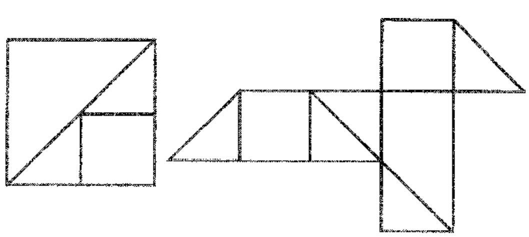
98. Varunahas lotsofpagesofsquarepaper,whereby eachpagehas anareaof4.She cuts eachofthepages into right-angledtriangles and squares (see the left hand diagram}. Shetakes afewof these pieces and'"forms theshapeintherighthanddiagram.Howbigistheareaofthisshape?

(1)4

(2)$\frac{9}{2}$

(3)5

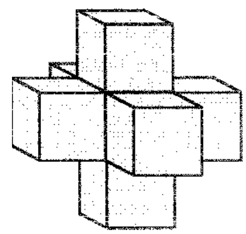
(4) 6



99. Abucket isfilled halfway with water. Acleaning liquid fillsanother 2litresofliquid i)ltOthebucket.Nowthe bucketis three-quarters full.Howmanylitresofwaterintotalcanfitinto thebucket?

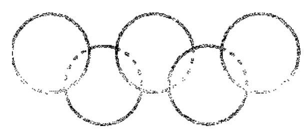
(1)10Litre (2)8Litre (3)6Litre (4)4Litre

100. Geet buildsthe sculpture shown fromseven cubeseach ofedge length 1.How many more ofthesecubesmust headdtothesculpturesothathebuildsalargecubeofedgelength3?



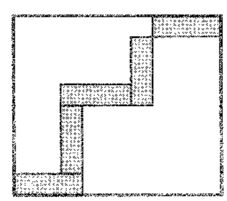
(1)12 (2)14 (3)16 (4)18

101. Five circleseachwithanareaof1cm^{2}overlapeachothertoformthefigureinthediagram.Thesectionswhere twocirclesoverlap,eachhaveanareaof$\frac{1}{8}$cm^2.Howbigisthearea,whichiscompletelycoveredbythe figureinthediagram?



(1) 4cm^2 (2) $\frac{9}{2}$cm^2 (3) $\frac{35}{8}$ cm^2 (4) $\frac{19}{4}$ cm^2

102. 5congruentrectangles are positionedin asquare with sidelength24asshown in the diagram. How big isthe area ofone ofthese rectangles?



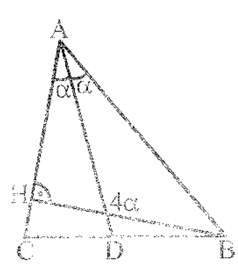
(1) 16cm^2

(2) 18cm^2

(3) 24cm^2

(4) 32cm^2

103. In triangleABC(seesketch)ADistheangle bisector oftheangleatAand BH isthe height fromsideAC.The obtuseanglebetweenBHandADis fourtimesthesizeofangle*LDAB*.Howbigistheangle*LCAB?*



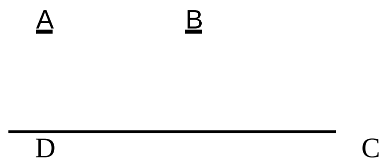
(1)30° (2)45° (3)60° (4)75°

104. Thesidesofarectangleare6cmand 11cmlong.Youselectoneofthelongsides.Thenthe anglebisectors oftheangles at the ends ofthis side are drawn.Theysplitthe opposite long side intothreepieces.Howlong arethesepieces?

(1) 2cm.7cm,2cm (2) 3cm,5cm,3cm

(3)4cm.3cm.4cm (4)5cm,1cm,5cm

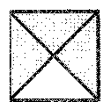
1os. ThequadrilateralABCDhasrightanglesonlyincomersAandD.Thenumbersinthediagramgivetherespective areasofthetrianglesinwhichtheyarelocated.HowbigistheareaofABCD?



(1)60 (2)45 (3)40 (4)35

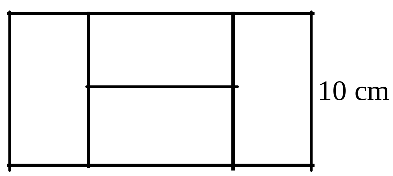
106. A 5x5 squareis coveredwith 1x1 tiles. The design oneach tile is madeup ofthree dark triangles and one light triangle (see diagram). The triangles of neighbouring tiles always have the same colour where they join along an edge. The border ofthelarge square ismade ofdark and light triangles.What isthesmallest number ofdark trianglesthat couldbe amongstthem?

(1)4 (2)5



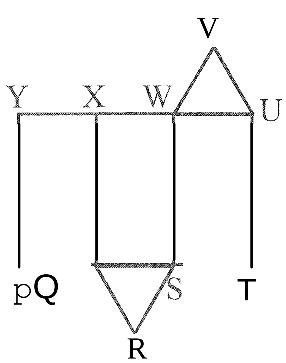
(3) 6 (4)7

107. Arectangleismadeof4equallysized,smallrectangles.Thesmallersidehassidelength10cm.Howlong is thelonger side?



(1)10cm (2)20cm (3)30cm (4)40cm

108. Thediagramshowsthenetof athree-sidedprism.Whichlineofthediagramformsanedgeof theprismtogether with line UV when thenet is foldedup?



(l)WB (2)XW (3) XY (4)QR

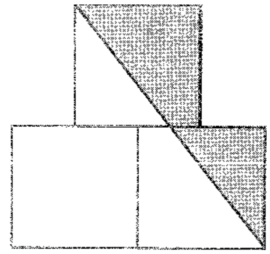
109. The side lengthsofa triangle are6,10and 11.Anequilateraltriangle hasthe same perimeter asthistriangle. How long is oneside ofthe equilateral triangle?

(1) 18 (2)11 (3)10 (4) 9

110. AcyclistcoversadistanceofSminonesec:on<l.Thewheelsofhisbikeeachhaveacircumferenceof 125cm.Howmanycompleteturnsdoeseachwheeldoin5seconds?

(1)4 (2)5 (3)10 **(4)**20

111. Thediagramconsists of three squareseach oneof side length 1. The midpoint ofthe topmost square isexactly abovethecommonsideofthetwoothersquares.Whatistheareaofthesectioncolouredgrey?

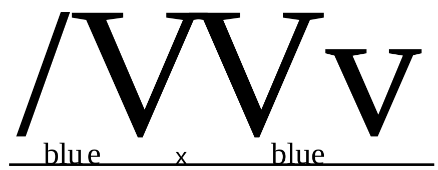


(1) $\frac{3}{4}$ (2) $\frac{7}{8}$ (3) 1 (4) $1\frac{1}{4}$

112. Duringathunder stormitrained15liters per squaremeter.Byhow muchdidthewater level ofanoutdoor swimming pool increase?

(1)150cm (2)0,15cm (3)15cm (4) 1,5cm

113. Eachsideofeachtriangleinthediagramispaintedeitherblue,greenorred.Fourofthesidesarealready painted.Whichcolour canthe linemarked,,x"have,ifeachtrianglemusthaveallsidesin differentcolours?

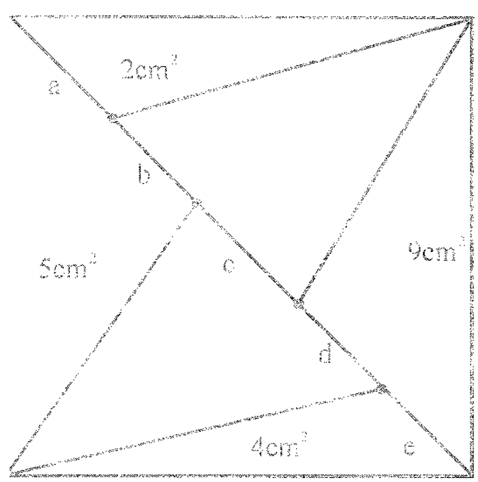


(1)onlygreen (2)onlyred (3)onlyblue (4)eitherredorblue

114. Rama addedthe lengths ofthree sides ofa rectangle and obtained44 cm. Krishna also added thelengths of threesidesofthesamerectangleandobtained40cm.Whatistheperimeteroftherectangle?

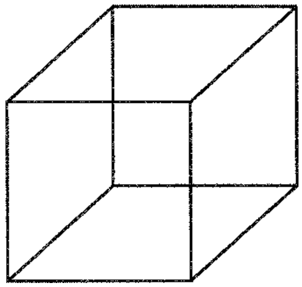
(1)42cm (2)56cm (3)64cm (4)84cm

115. A square with area 30is split into two by its diagonal and then Splitinto triangles as shownin the diagram. Some ofthe areas ofthe triangles are given in the diagram. Which ofthe line segments a, b, c, d, e ofthe diagonal is the longest?



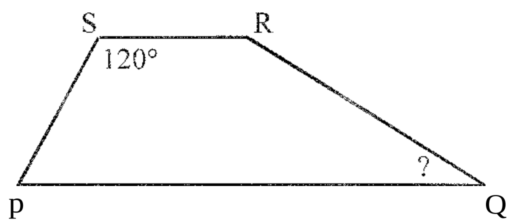
(1)a (2)6 (3)c (4)d

116. Shivajihassevenpiecesofwireoflengths1cm,2cm,3cm,4cm,5cm,6cmand7cm.Heusessome ofthose piecesto form awiremodel ot acubewith sidelength 1.Hedoes notwant anyoverlapping wire parts. Whatis thesmallestnumberofwirepiecesthathecanuse?



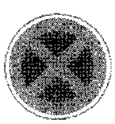
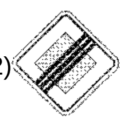
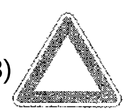
(1)1 (2)2 (3)3 (4)4

117. InthetrapeziumPQRSthesidesPQandSRareparallel.Also∠RSP= 120°and$\overline{RS} = \overline{SP} = \frac{1}{3}\overline{PQ}$. Whatis thesizeofangle ∠PQR?

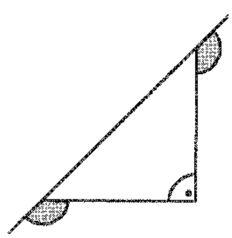


(1)15° (2)22.5° (3)25° (4)30°

118. Whichoftheroadsignshasthemostaxesofsymmetry?

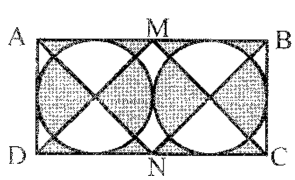
(1)  (2)  (3)  (4)

119. Whatisthesumofthetwomarkedangles?



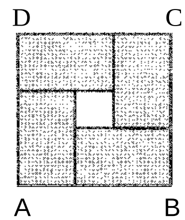
(1)150° (2)180° (3)270° (4)320°

120. IntherectangleABCDthesideADis10cmlong.MandNarethemidpointsof thesidesABandCDrespectively. How big is thegrey area?



(1)50cm^2 (2)80cm^2 (3)100cm^2 (4)120cm^2

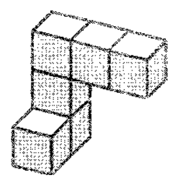
121. WithinthesquareABCDtherearefouridenticalrectangles(seediagram).

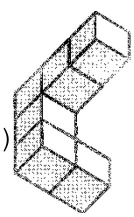
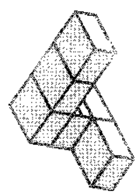
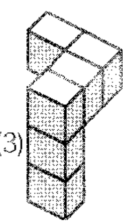
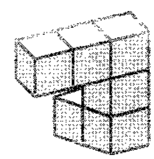


Theperimeterofeachrectangleis16cm.Whatistheperimeterofthissquare?

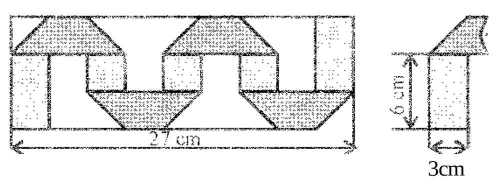
(1)20cm (2)24cm (3)28cm (4)32cm

122. Rohini hasgluedtogethersomecubesandhasobtainedthesolidshownontheright.She turnsitaround*to* checkitoutfromdifferentsides.Whichviewcanshenotobtain?



(1)  (2)  (3)  (4) 

123. A 3cm wide strip ofpaper isdark on one side and light onthe other. The foldedstrip ofpaper lies exactly withinarectanglewithlength27cmandwidth9cm(seediagram).Howlongisthestrip ofpaper?

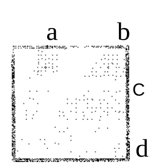


(1)36cm (2)48cm (3)54cm (4)57cm

124. Sevenidenticaldice(eachwith1,2, 3, 4,5and6 pointson theirfaces)areglued togethertoformthesolid shown. Facesthat areglued together each have thesame numberofpoints. How many pointscan beseen on the surface ofthe solid?

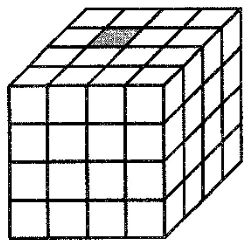
(1)24 (2)90 (3)95 (4)105

125. Ina squarewitharea36thereare greypartsas shownin thediagram.Thesumofthe areas ofallgreyparts is27.Howlongarethedistancesa,6,c anddtogether?



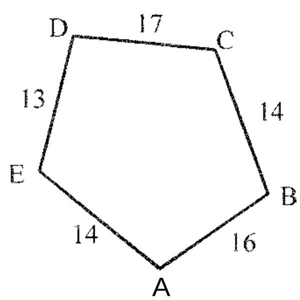
(1)4 (2)6 (3) 8 (4)9

126. Abigcubeismadeupof64smallcubes.Exactlyoneofthesecubesisgrey(seediagram).Twocubesareneighbours iftheyshareacommonface.Ondayonethe greycubecolours all itsneighbouringcubesgrey.Ondaytwo all greycubes again colour all their neighbouringcubes grey. How many ofthe64little cubes are grey at the end of the second day?



(1)13 (2)15 (3)16 (4)17

127. Thediagramshowsapentagonandindicatesthelengthofeachside.Fivecirclesare drawnwithcentresA, B, C, Dand E. On each side ofthepentagonthe two circles that aredrawn around the endsofthat side touch eachother.Whichpointis thecentre ofthebiggestcircle?

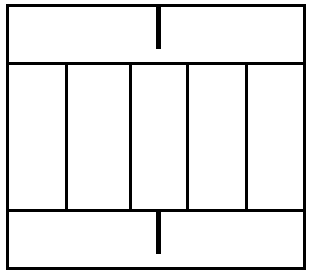


(1)A (2) B (3) C (4)D

128. Atriangle ABChasside lengths 6 cm,10cm and11 cm. An equilateraltriangle XYZhas thesameperimeter as the triangleABC.Whatare thesidelengthsofthetriangleXYZ?

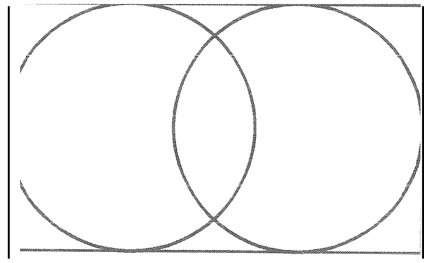
(1)6cm (2)9cm (3)10cm (4)11cm

129. Alarge rectangle ismade upof9 equally big rectangles. Thelongersideof eachsmall rectangle is10cmlong. Whatistheperimeterofthe largerectangle?



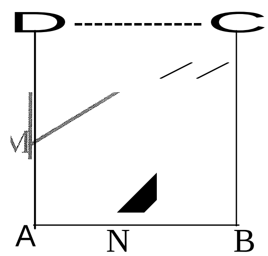
(1)40cm (2)48cm (3)76cm (4)81cm

130. Two circlesareinscribedinto an11cmlongand7 cm widerectangleso thatthey eachtouchthreesidesof therectangle.Howbigis thedistancebetweenthe centresofthetwo circles?



(1)1cm (2)2cm (3)3cm (4)4cm

13L The squareABCD hasside length3 cm. Thepoints MandN, which lie on the sides ADand AB respectively, arejoinedto thecornerC.Thatwaythe squareis splitup intothreeparts withequalarea.How longis the line segmentOM?

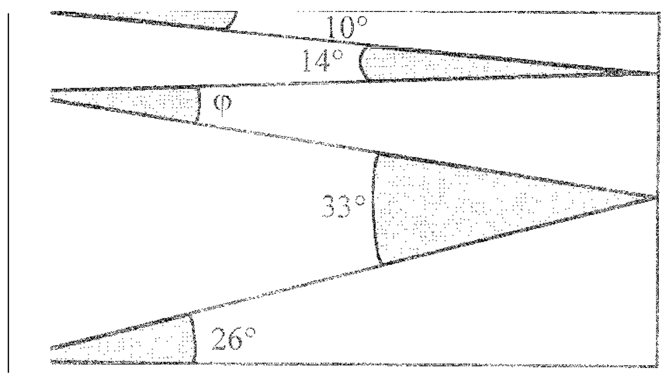


(1)0.5cm (2)1cm (3)1.5crn **(4)**2crn

132. A rectangle is split up into 40 equally big squares. The rectangle consists of more than one row of squares. Andreascoloursinall squares ofthemiddlerow.Howmanysquares didhenotcolourin?

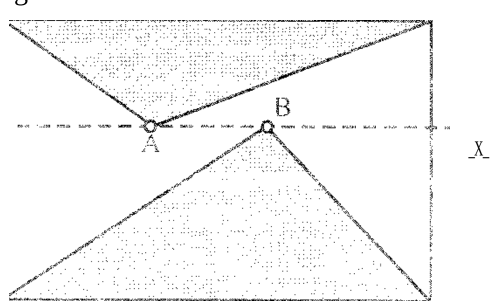
(1)20 (2)30 (3)32 (4)35

133. Sudhirdraws a zig-zagline inside arectangleas shownin the diagram.For thatheuses the angles 10°. 14°, 33°and26°.How big is anglecp?



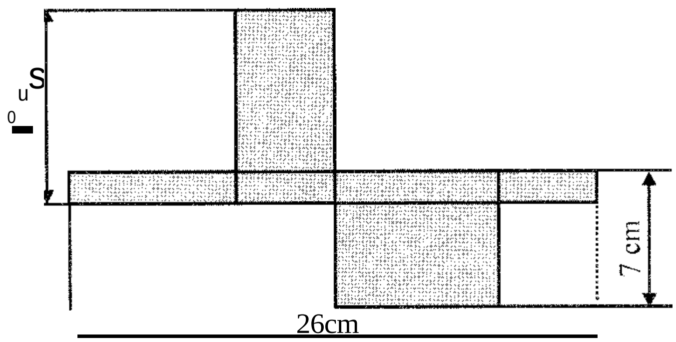
(1)11° (2)12° (3) 16° (4)17°

134. Thediagram showsarectangle andastraight iinex, which ispara!leltooneofthesidesofthe fectangle. There a1etwopointsAandBonxinsidetherectangle.The sumoftheareasofthetwotrianglesshadedingrey is 10 err?. Ho<vvbig is the area ofthe rectangle?



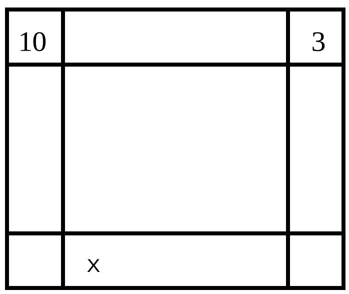
(2)20crn2 (3)22crn2

135. Thediagramshowsthenetofaboxconsistingonlyofrectangles.Howbigisthevolumeofthebox?



(1)43cm^3 (2)70cm^3 (3)80cm^3 (4)100cm^3

136. Rita wants towrite anumberinto everysquare ofthediagramshown.Everynumbershouldbe equal to the sum ofthe twonumbers from the adjacentsquares. Squares areadjacent ifthey share oneedge. Two numbers arealreadygiven.Whichnumberis shegoingtowriteintothesquaremarkedwithx?

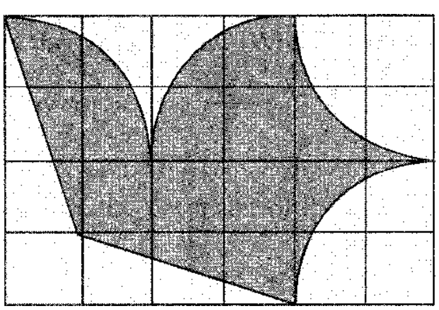


(1)10 (2) 7 (3)13 (4)-13

137. Rohit runsalong theedge rounda 50mlong rectangular swimmingpool, whileat the sametimeSachin swims lengthsinthepool.Rohit runsthreetimesasfastasSachin swims.WhileSachinswims6lengths, Rohitmanages 5rounds aroundthepool.How wideis the swimmingpool?

(1)25m (2) 40m (3)50m (4)80m

138.

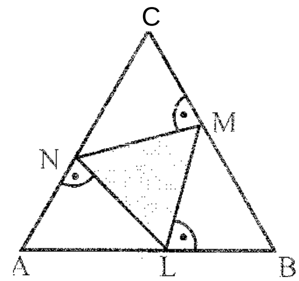


Allen aviationclubdesigns a flag with a flying "dove" ona 4 x6-grid. The area ofthe"dove" is **192** cm2.The perimeterofthe,,dove"ismadeupofstraightlinesandcirculararcs.Whatmeasurementsdoesthe flaghave?

(1) 6cmx4cm (2)12cmx8cm

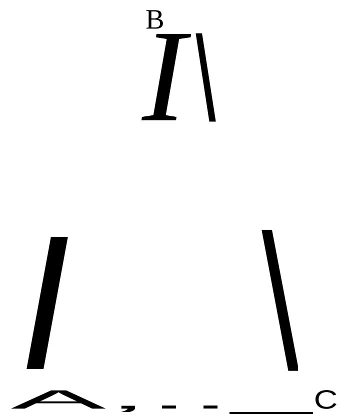
(3)20cmx12cm (4)24cmx16cm

139. ThepointsN,MandLlieonthesidesofanequilateraltriangleABCsothatNM*1*-BC,ML1-ABand LN*1*-ACholdstrue.TheareaoflhetriangleABCis36cm2.WhatistheareaofthetriangleLMN?



(1)9cm2 (2)12 cm2 (3)15cm2 (4)16cm2

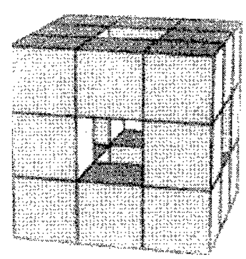
140. Intheisoscelestriangle ABC(withbaseAC) the pointsKa.nd Lareadded on thesidesABand BCrespectively so that AK KL =LB and KB = AC.How big :s the angle ABC?



(1)30° (2)35° (3)36° (4)40°

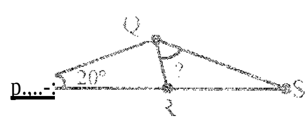
14LA3x3x3cubeismadeupofsc1all1x**1**x1cubes.Thenthemiddlecubesfromfronttoback,fromtop to bottomandfromrighttoleftareremoved(see diagram)

Howmany1x1x] cubesrernam?



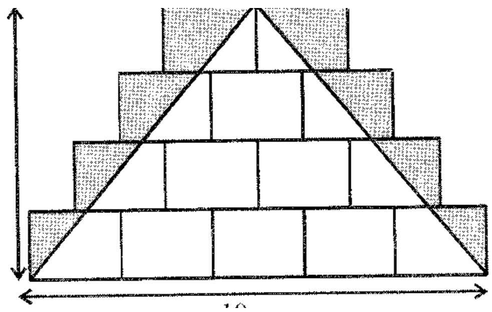
1S (2)18 (3)20 (4)21

142. The follow,ng i:1forrnab:m is known about triangle PSQ: LQPS 20° The +rians11e PSQ has upmto two smaller triani.:;les the !:ne QR as ::;hown. Itis k,iown tlv1t PQ PR QSHow bi(3i-;tk:angleRQS?



(1)50° (2)60° (3)65° **(4.)70°**

143. Mishadrawssomecongruentrectanglesandonetriangle.Shethenshadesin greythosepartsoftherectangles thatlieoutsidethetriangle(seediagram).Howbig1stheresultinggreyarea?

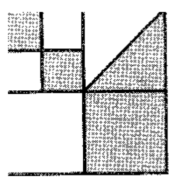


(1)10cm2 (2)12cm2 (3)14cm2 (4)15cm2

144. Inwhichoftheregularpolygonsbelowisthemarkedanglethelargest?

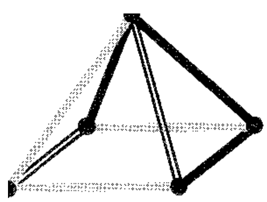
(1)O 1210 (3)O (4)

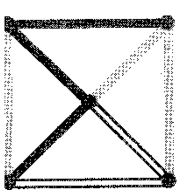
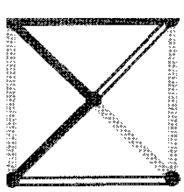
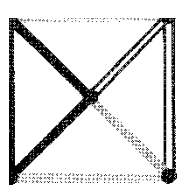
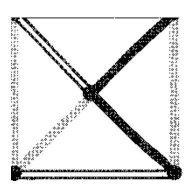
145. Alarge squareisdivided intosmaller squares. Inoneofthesquaresadiagonal isalsodrawn. What fraction ofthe large square is shaded?



(1) $\frac{3}{8}$ (2) $\frac{4}{9}$ (3) $\frac{1}{3}$ (4) $\frac{1}{2}$

146. Whichofthefollowingshowswhatyouwouldseewhentheobjectinthediagramisviewedfromabove?



(1)  (2)  (3)  (4) 

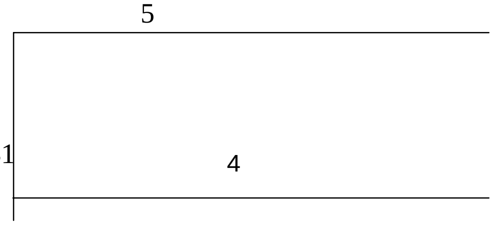
147. Sacha'sgardenhastheshapeshown.Allthesidesareeitherparallelorperpendiculartoeachother.Some ofthedimensionsareshowninthediagram.Whatis theperimeterofSacha's garden?

(1)22

(2)23

(3)24

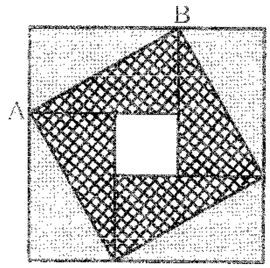
(4)25



148. Adhyatmbuys 27identicalsmall cubes,eachwithtwoadjacentfacespaintedred. Hethenusesall ofthese cubes tobuiidalargecube.Whatisthelargestnurnberofcomplerslyredfacesc,fthelargecubethathecanrnake?

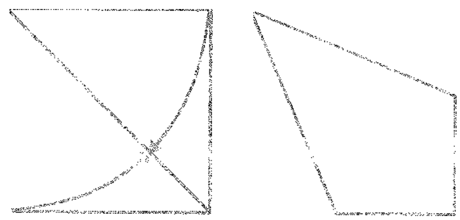
(i)2 (2)3 (3)4 (4)5

149. Alarge square consists ofburidenticalrectangles and a smalisquare.The area ofthe large square is 49cm2andthelengthofthe diagonalAB ofoneoftherectanglesis 5cm.Whatistheareaofthe small,;qucirc?



(1)lcm2 (2)4cm2 (3)9cm2 (4\J6

l50,,Ranvir tookasquare piece ofpaperand folded *tvJo* ofitssides tothediugonaJ,asshovJn,to obtainaquadrilateral Whatis thesizeofthelargestangleof 1.hequadrilateral'?



(1)112s· (2)1120° (3)125° (4)135°

15L Saniyawritesapositiveintegerone3.chedgeofa5quare.Shealsowritesateachvei-i"extheproductofthe numberson thetwoedgesthat meetatthatvertex Thesumofthenurnbersatth'"verticesis15.Whatisthe sum ofthenumbers on the edgesofthesquare?

(1)6 (2)7 (3) 8 (4)10

152. Sophia has52 identicai isosceles righh::ingledtriangles. She 1.vamstomakeasquare usingson,eof them.How many differentsizedsquares can shernake'?

(1)6 (2)7 (3) 8 (4) 9

l53. Fourch.ildrenareinthefourcornersoJa10rn2bn1.pool.'T'heirtrainerisstandingsornev.Jhereononeside ofthepooi.Whenhecallsthem,threechildrengetoutandwalkasshortcdis1anceaspos,ibleroundthepooi tomeethim.TheywalkSOrnintotal.Whalis theshortestdistancethetrainerneedstowalktc*ge1*tothe fourth child?

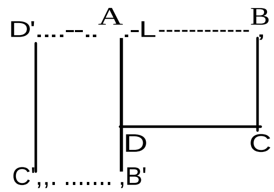
(1)10rn (2)J.2m (3)15m (4)20ff,

SECTION B

154. ABCisarightangledtriar:g!ewithB 909.BDEFisasquare.BEisperpendicuiartoAC.Themeasureof /DECis ________

155. Fron1apointwithinanequilateral1ri,mgieperper;dicularsaredrawntothethreesidssandare*b*,7ann9crns is length.The cAthetrianqieis cm.

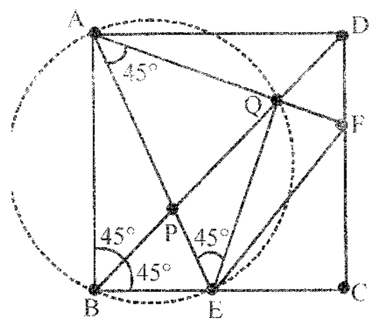
156. ABCDisarectangle rotatedclockwiseabout Aby90Qasshown.TherotationtakesBtoB',CtoC',DtoD'. AB=6cm,BC'=l0cm.ThebreadthoftherectangleABCDis______________ _



157. ABisalinesegment2000cmlong.Thefollowing designof semicircleisdrawnonAB, withAP= 5cmand repeating the designs. The area enclosedby the semicircular designs from A to Bis______________ _

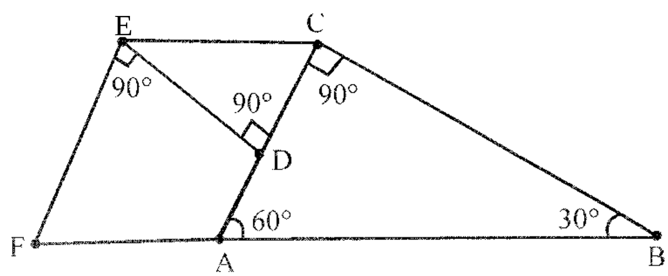
158. ABCD isasquare.EandFarepointsrespectivelyonBCandCDsuch that*LEAF*=45°.AEandAFcut the

diagonalBOatP,Q respectively. Then findvalue of $\frac{\text{AreaofAEF}}{\text{AreaofAPQ}}$

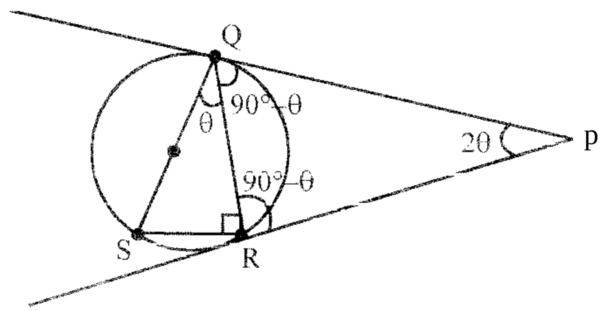


159. IntheadjoiningfigureBACisa30°-60°-90°trianglewithAB=20.DisthemidpointofAC.Theperpendicular at Dto AC meetsthe line paralleltoABthrough Cat E.Theline through E perpendiculartoDEmeets BA producedatF.

If OF =*S* findx.



160. PRandPQaretangentstoacircleandQS isadiameter.Find--- $\frac{\text{L'.'.'.QPR}}{\text{L'.'.'.RQS}}$

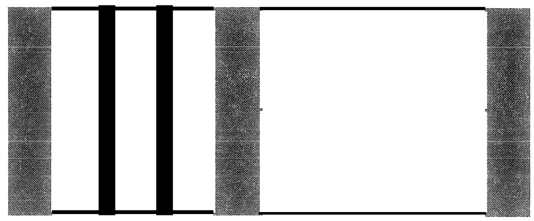


161. ABCD is atrapeziumsuchthat AB IICD. AB=10cm;BC=8cm;CD=DA=5cm.Findtheheightof the trapeziumand also find AC and BO.

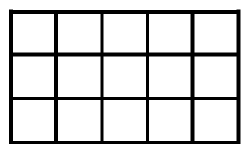
162. ABCD is a convex pentagon with*LA=LB=LD*= 90°;*LC*= *LE*.Sides AB, BC, CD, DE and EA are extended to K, L, M, Nand Orespectively. Theexterior angles*LOAB*, LKBC, LLCD, LMDEand LNEA arebisectedandthebisectorsareproduced ineitherdirectiontoformthepentagon PQRST.Findtheangles of this pentagon.

163. AllsidesoftheconvexpentagonABCDEareequalinlength*LA*=*LB*=90°.Thenfindthevalueof*LE*.

164. Abarcodeisformedusing25 blackandcertainwhitebars.Whiteandblackbarsalternate.Thefirstandthe lastareblackbars.Someoftheblackbarsarethinandothersarewide.Thenumberof whitebarsis15more than the thin black bars. Then find the number ofthick black bars.



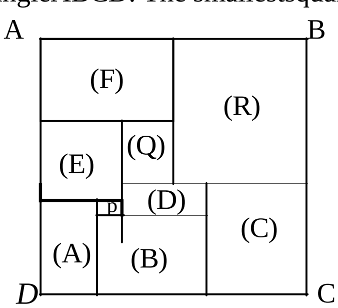
165. Bydrawing 10linesofwhich4arehorizontal and 6 are vertical crossingeach other asin the figure, onecan get15 cells.Withthesame10 linesof which 3areverticaland 7horizontalweget12cells. Find the maximum numberofcells that couldbe gotby drawing20lines (somehorizontalandsomevertical)is



166. Theangleofapolygonareintheratio2:4:5:6:6:7.Calculatethedifferencebetweenthegreatest and least angle ofthe polygon.

167. Theperimeterofarightangledtriangleis 132.Thesum ofthesquareofallits sidesis6050.Findthesum ofthe legs ofthe triangle.

168. Nine squares are arrangedto form a rectangleABCD. The smallestsquareP has anarea 4sq.units.Findthe areas ofQ and R.



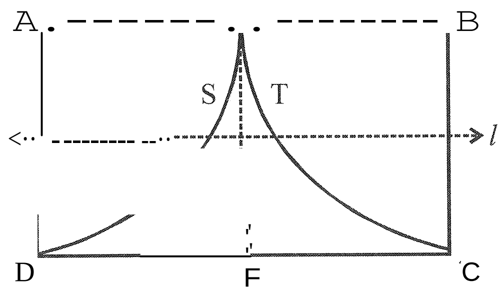
169. ABCD is a trapeziumwith ABand CD parallel. If AB= 16cm, BC=17cm, CD=8 cm, DA=15cm then theareaofthe trapezium(incm^2)will be ?

170. Asquareis inscribedinanothersquareeachofwhosefourverticeslinesoneachsideofthesquare.Thearea ofthesmallersquareis$\frac{25}{49}$timestheareaofthebiggerone.Thencalculatetheratiowithwhicheachvertex ofthesmallersquaredividesthesideofthebiggersquare.

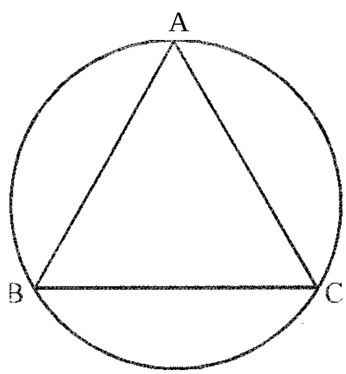
171. PSR isanisoscelestriangle inwhich PS= PR.SPisproduced to Osuch thatPO =SP.Then thevalueofLSRO is equal to?

112.CABisananglewhosemeasureis70°.ACFGand ABDEaresquaresdrawnoutside theangle.The diagonal FAmeetsBEatH.ThencalculatethemeasureoftheL'.'.EAH.

173. InrectangleABCD,AB=2BC= 4 cm EandFare midpoints ofABandCD respectively.ESD and ETC are arcs ofcirclescentredatA andBrespectively.Iftheperpendicularbisectorline*l*ofEF cuts the arcsat S and T asinthediagram,thenSTis equalto(incm)

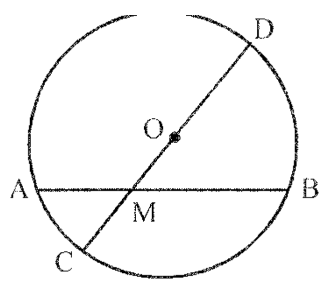


174. Triangle ABC is equilateral of side length 8 cm. Each arc shown in thediagram is an arc of a circle with the oppositevertexofthetriangleasitscentre.Calculatethetotalareaenclosedwithintheentirefigureshown (in cm^2).



175. ABCDisasquare.AlineAXmeetsthediagonalBOatXandAX=2016cm.Thelengthof CX (in cm) is equal to ?

176. 0is the centreofacircle ofradius15cm.Misapointata distanceof5cm from0.AMBis anychordof the circle throughM,then calculatethevalueofAMxMB.

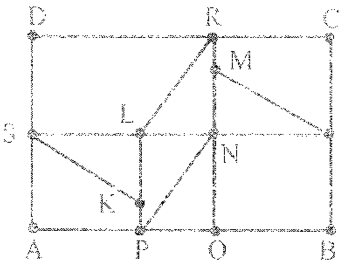


177. An isosceles trapezoidiscircumscribedabout a circle ofradius 2cm and the area ofthetrapezoidis $20cm^2$.Find the length ofequal sides ofthe trapezoid.

178. Atrianglehassideswithlengths13cm, 14cm,and15cm.Acirclewhosecentreliesonthelongest sidetouches the other two sides.Find the radius ofthe circle (in cm).

179. ABC ,m<l ADE uri! lwu :;ec,.mlsufu circleuf 1c.1Jiu::s3uu. /\.i::sul u lb w1ceuf5 cm from the centre oflh circle.Thesecantsincludeanangleof30°.TheareaoftheL1ACEis$10cm^2$.Thencalculatetheareaofthe ,\ADB(incm^2).

180. In the figure, ABCD is a rectangle with sides 24 units, 32 units. SKL, PLN, QNM, RNL are congruent right angled triangles, then calculate the area of each triangle.



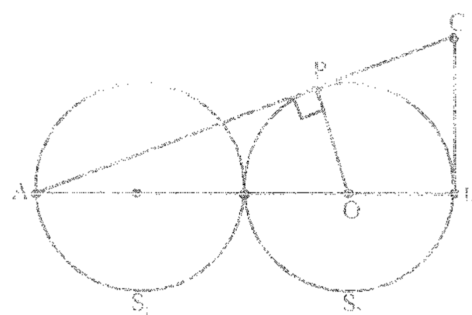
181. AB and AC are two straight line segments enclosing an angle 70°. Squares ABDE and ACFG are drawn outside the angle BAC. The diagonal FA is produced to meet the diagonal EB in H, calculate the value of ∠EAH.

182. In the pentagon ABCDE, ∠D = 2 ∠B and the other angles are each equal to half the sum of the angles ∠B and ∠D. What will be the largest interior angle of the pentagon.

183. A triangle whose sides are integers has a perimeter 8. The area of the triangle is.

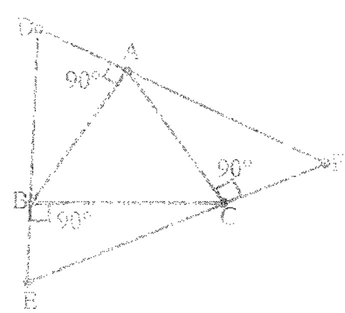
184. ABC and ADC are isosceles triangles with AB = AC = AD. ∠BAC = 40°, ∠CAD = 70°. Calculate the value of ∠BCD : ∠BDC

185. In the figure below two equal circles S_1, S_2 of radii 2 units each touch each other. AB is the common diameter. The tangent at B meets the tangent from A to the circle S_2 at C as shown. If $BC = K\sqrt{2}$ then the value of K is.



186. In the figure below, ΔABC is equilateral AD, BE and CF are respectively perpendicular to AB, BC and AC.

Calculate $\dfrac{\text{Area of } \Delta DEF}{\text{Area of } \Delta ABC}$



187. ABCD is a parallelogram. P is a point on AD such that $\dfrac{AP}{AD} = \dfrac{1}{2015}$. Q is the point of intersection of AC and BP. Calculate $\dfrac{AQ}{AC}$.